

Location / Section	Technical Issue Observed	Comment to Incorporate	Suggested Correction / Way Forward
Cover page	The title is acceptable but can be made more technically precise.	The title should reflect the actual contribution: PPO/MAPPO-based adaptive traffic control for the Palapye A1 corridor.	Suggested title: “Adaptive Multi-Agent Reinforcement Learning for Traffic Signal Optimization along the Palapye A1 Urban Corridor.”
Cover page	Formatting issues: “Supervisor : Dr Bokamoso Basutli :” has an extra colon.	The cover page needs professional formatting.	Change to Supervisor: Dr. Bokamoso Basutli . Ensure spacing is consistent for name, ID, signatures, programme, and date.
Table of contents	The structure shows Chapters 1–9, but Chapter 5 uses “Discussions” and subsections appear as 5.0.1, 5.0.2, etc.	Chapter numbering is technically weak and must be corrected.	Rename “Discussions” to “Discussion” and make subsections 5.1, 5.2, 5.3 , etc., not 5.0.1 .
Overall structure	The report contains Semester 1 PPO work and Semester 2 MAPPO work, but the transition is not clean enough.	The project has two phases, but the report must explicitly separate them.	Add a short section before Chapter 6: “Scope Extension from Single-Agent PPO to Multi-Agent MAPPO.”
Chapter 1 Introduction	The introduction is clear and locally relevant, but it uses broad claims such as “future-proof Palapye’s transport network.”	Avoid overclaiming. This is a prototype/simulation study, not a deployed municipal system.	Replace with: “provide a simulation-based foundation for future adaptive traffic control deployment in Palapye.”

Background	Palapye's A1 corridor importance is well explained, including BIUST, Morupule Coal Mine, and regional transit relevance.	This is a strong contextual justification.	Strengthen it by adding a small map or corridor diagram showing the three selected intersections.
Figure 1.1	The traffic image from Mmegi Online is useful but only illustrative.	A real project should not rely only on a media image for traffic justification.	Add a statement that the image is contextual only and include actual field observations, traffic counts, or Google Maps screenshots if available.
Problem Statement	The problem statement is too short.	It states the issue but does not mention the specific technical gap: fixed-time control lacks sensing, prediction, and multi-intersection coordination.	Expand to include: "Existing fixed-time control does not use real-time queue lengths, corridor-level coordination, or adaptive phase selection under varying demand."
Objectives	Objectives are now aligned with the work: YOLO, SUMO corridor modelling, MAPPO, and comparative evaluation.	Good objective structure, but the first objective may overstate implementation because YOLO is tested independently, not integrated into the SUMO control loop.	Revise Objective 1 to say: "develop and test a YOLOv8-based vehicle detection module as a sensing foundation for future integration."
Contributions	The contribution claims "first application" in Botswana.	This may be true, but it needs careful wording unless verified through literature.	Write: "To the best of the author's knowledge, this represents one of the first reported applications..."

Contributions	The report claims a “calibrated simulation environment.”	Calibration requires real traffic counts or measured signal timing data. The report itself states that no historical real data were available.	Replace “ calibrated simulation environment ” with “ synthetically parameterised SUMO simulation environment. ”
Chapter 2 Literature Review	The literature review is relevant and covers sensors, fixed-time, actuated, adaptive, MPC, DTA, MILP, RL, PPO, and MARL.	The breadth is good, but the review sometimes reads like a textbook instead of a critical review.	Add a summary comparison table: method, required data, scalability, real-time suitability, limitation, relevance to Palapye.
Section 2.1 Sensory Circuits	The heading “Sensory Circuits” is not the best technical heading.	This section discusses vehicle sensing technologies, not electrical circuits only.	Rename to “ Traffic Sensing and Vehicle Detection Technologies. ”
Figure 2.1	“Block Diagram of The System” is too generic and appears before enough explanation.	The figure does not clearly represent the implemented system.	Replace it with a clearer pipeline: sensing → traffic state extraction → controller decision → SUMO/traffic signal actuation → performance metrics.
Camera vision review	Classical OpenCV detection is discussed, followed by YOLOv8.	Good comparison, but the report should explain why YOLOv8 was selected over classical OpenCV.	Add a justification table: OpenCV-MOG2 vs YOLOv8 for lighting robustness, occlusion, accuracy, computational cost, and queue estimation.
YOLOv8 discussion	The report states GPU YOLOv8 can exceed 100 FPS on mid-range hardware.	This may be too general unless hardware is specified.	Add actual test hardware, FPS achieved, model version used, confidence threshold, and sample video resolution.

LiDAR and radar sections	These sections are technically useful but not directly used in the implementation.	Keep them concise and explain why camera vision was selected.	Add a closing paragraph: “Camera vision was selected due to lower cost, easier deployment, and availability of sample footage.”
Traffic flow fundamentals	Flow, density, speed, saturation flow, Webster’s formula, and LOS are appropriate.	This section is strong, but the equations must be connected to the selected metrics.	After LOS, add: “In this project, waiting time and queue length serve as simulation proxies for control delay and LOS.”
RL update rule	Q-learning is explained, but the implemented models are PPO and MAPPO.	Q-learning is useful background, but PPO/MAPPO deserve more emphasis.	Reduce Q-learning detail slightly and expand PPO objective, advantage estimation, clipping, entropy, and actor–critic structure.
MARL section	The CTDE explanation is one of the stronger parts of the report.	The explanation is relevant and technically mature.	Keep it, but add a small table comparing IQL, MADDPG, QMIX, MAPPO specifically for traffic signal control.
Chapter 3 Methodology	The methodology states the workflow includes DRL, ML, YOLO, SUMO, and TraCI.	Good structure, but the wording still mixes “training dataset” and “online RL interaction.”	Clarify that PPO/MAPPO is trained by interaction with SUMO, not by ordinary supervised dataset training.
Section 3.1	The report says “Dataset training using SUMO-generated synthetic traffic scenarios.”	In RL, the agent does not simply train on a static dataset; it learns through state–action–reward interaction.	Replace with: “SUMO-based environment generation and RL interaction for policy learning.”
YOLO integration	The report clearly admits YOLO was tested independently and not	However, the objectives/contributions must not make it appear fully integrated.	Use consistent wording everywhere: “YOLOv8 module was independently

	integrated into SUMO. This is honest and good.		validated; full closed-loop integration is future work.”
Figure 3.1	The methodology diagram is useful and clearly shows CameraVision not integrated into SUMO.	This is a strong visual because it prevents overclaiming.	Keep the dashed “future work” link, but add TraCI explicitly between SUMO and PPO/MAPPO.
Figure 3.2	The architecture shows three modules but still centres PPO.	Since later chapters focus on MAPPO, the architecture should include both Semester 1 and Semester 2 designs.	Add a second architecture diagram for MAPPO: three SUMO intersections → local observations → shared actor → global critic during training → decentralised phase actions.
SUMO scenario generation	The timing descriptions such as “0–10s, 10–30s, 30–50s, 50–70s” are unrealistic for real traffic demand periods.	Traffic scenarios should not be described as only 10–20 seconds unless these are simulation labels.	Convert to scenario phases or episodes, e.g., low demand, moderate demand, peak demand , and state the actual simulation duration.
Dataset generation	It states that no real historical traffic data existed.	This is acceptable, but synthetic data must be justified and bounded.	Add assumptions: arrival rates, lane geometry, turning probabilities, vehicle types, route probabilities, and speed distributions.
Figure 3.3	The SUMO network screenshot is useful but not enough.	It does not show lane numbers, intersections, or demand flows clearly.	Add a labelled network diagram showing TL_A, TL_B, TL_C , entry edges, exit edges, lane counts, and turning movements.

Section 3.2.2 Real-Time Data Collection	The YOLO screenshot shows detections, queue labels, and traffic metrics.	This is good practical evidence, but it lacks quantitative validation.	Add a validation table: manual count, YOLO count, detection accuracy, false positives, false negatives, FPS, queue estimation error.
CameraVision	The report mentions centroid tracking and optical flow.	The exact algorithm is unclear.	State whether the implementation uses centroid tracking, SORT, DeepSORT, optical flow, or a custom tracker. Include confidence and IoU thresholds.
Section 3.2.3 Model Control	The model-control section begins abruptly: "The decision-making component of the system is a Deep..."	This indicates a writing or formatting break.	Rewrite the opening paragraph and define the exact PPO environment: state, action, reward, step interval, phase constraints, and termination condition.
Chapter 4 Results	Chapter 4 evaluates single-intersection PPO versus fixed-time.	This is useful, but should be labelled as Semester 1 baseline results.	Rename Chapter 4 to "Single-Intersection PPO Results and Analysis."
Performance metrics	The five metrics are appropriate: delay, queue length, throughput, stop ratio, and pressure.	Good metric selection, but units and definitions must be consistent.	For each metric, provide formula, unit, and whether lower/higher is better.
Table 4.1 / Table 5.1 duplication	The same performance tables appear in Chapter 4 and Chapter 5.	This creates unnecessary repetition and weakens report organization.	Keep the numerical tables in Chapter 4 and use Chapter 5 for interpretation only, or merge Chapters 4 and 5.

Statistical testing	The report uses p-values, Mann–Whitney U, and Cohen’s d.	This is strong, but it must explain sample size and independence.	Add: number of episodes, number of timesteps, whether samples are independent, and why Mann–Whitney U was selected.
PPO aggregate delay	The conclusion admits PPO had a 10.1% aggregate delay disadvantage, although it improved throughput, queue length, and pressure.	This must be discussed more carefully, not hidden.	Explain that PPO may trade short-term delay for throughput and pressure balancing; then show phase-specific delay benefits during heavy/peak traffic.
Stop ratio	Fixed-time appears slightly better in aggregate stop ratio, but PPO improves during congested phases.	The interpretation is acceptable, but it needs clearer engineering framing.	State: “Aggregate stop ratio alone is misleading because it masks high-demand performance.”
Traffic pressure	PPO has higher pressure during peak/saturation in Table 5.6, but the text says this is strategic queue accumulation.	This needs more careful explanation because high pressure can also indicate poor control.	Add a plot showing pressure dissipation after peak demand. If pressure rises but clears faster, the claim is defensible.
Chapter 5 Discussion	The chapter repeats many Chapter 4 tables.	Discussion should interpret, not duplicate.	Convert Chapter 5 into a concise discussion with subsections: PPO benefits, delay trade-off, queue management, pressure behaviour, limitations.
Chapter 6 MAPPO Design	This is the strongest technical section. It clearly explains	The progression is technically logical.	Move part of this explanation earlier into Methodology so that the reader knows the

	moving from isolated PPO to corridor-level MAPPO.		final system is MAPPO before reaching Chapter 6.
CTDE architecture	The shared actor and centralised critic idea is well explained.	Good technical choice for three similar intersections.	Add mathematical notation for global state (s_t), local observation (o_i), local action (a_i), joint reward (r_t), and shared policy ($\pi_{\theta}(a_i$
Observation space	The report states 22 dimensions per agent and 66-dimensional joint state.	This is good, but the exact 22 features must be fully justified.	Add a table explaining each feature, source in SUMO/TraCI, unit, range, and normalization method.
Action space	Three green phases are used.	The phase definitions need to be explicit.	Add phase table: action 0 = north-south green, action 1 = east-west green, action 2 = turning/secondary phase , etc.
Reward function	The global cooperative reward is appropriate.	However, the reward equation must be fully visible and justified.	Include exact reward formula and explain weights for waiting time, queue length, pressure, phase switching penalty, and throughput reward.
Green wave pre-computation	This is an interesting addition.	It must be clear whether the green wave is only used as a feature, as reward shaping, or as a baseline.	Add a short paragraph: “Green-wave pre-computation is not hard-coded control; it is used only for...”
Curriculum learning	Curriculum over demand scenarios is appropriate.	Good idea, but it needs evidence that curriculum improves stability.	Add a comparison: MAPPO with curriculum vs MAPPO without curriculum, even if brief.
Chapter 7 Training	MAPPO training over 1.5 million steps is credible.	Good, but reproducibility requires more details.	Add random seeds, number of parallel environments, rollout length, batch size, learning rate, gamma, lambda, entropy

			coefficient, clip range, and training hardware.
Figure 7.1	Training curves are useful and show convergence.	The text explains reward improvement and critic stabilization well.	Add a statement on whether training was repeated over multiple seeds or only one run.
Table 7.2	Qualitative comparison of fixed-time, SA-PPO, and MAPPO is useful.	However, qualitative comparison should be supported by quantitative values.	Add a quantitative comparison table: average waiting time, queue length, throughput, pressure, across all scenarios.
MAPPO vs single-agent PPO	The report explains that MAPPO prevents spillback better than single-agent PPO.	This is a strong technical argument.	Add one figure or table showing downstream queue/spillback behaviour for SA-PPO vs MAPPO.
Figure 7.2	MAPPO is compared against fixed-time and random controllers over six demand scenarios.	This is a strong result figure, but exact numerical values are not visible in the text.	Add a table with exact mean $\pm$ standard deviation for each scenario and controller.
Evaluation results	The report states MAPPO reduces rush-hour waiting time by about 60% and queue length by over 50%.	Strong claim, but must be supported numerically.	Add exact values: fixed-time waiting time, MAPPO waiting time, percentage reduction, p-value or confidence interval.
Random controller baseline	Random controller is useful but weak as a baseline.	The main baseline should remain fixed-time and single-agent PPO.	Add Webster fixed-time or actuated heuristic as an additional baseline if possible.

Conclusion	The conclusion clearly separates Semester 1 and Semester 2.	This is good, but the wording still needs caution because MAPPO is simulation-based.	Add: “The findings are simulation-based and require field calibration before practical deployment.”
Conclusion	It reports single-agent PPO results well: 48.5% throughput improvement, 27.8% queue reduction, and 27.3% pressure reduction.	These are strong results, but should be tied to the exact simulation duration and setup.	Mention: number of episodes, simulation timesteps, and demand phases in the conclusion.
Recommendations	Recommendations should be more technical and implementation-focused.	General recommendations are not enough for a strong FYP report.	Add: field traffic count collection, camera calibration, real-time YOLO integration, actuated baseline, safety constraints, pedestrian phases, emergency vehicle priority, and hardware deployment plan.
Appendix	Appendix only shows PPO training script.	Since MAPPO is the final contribution, the MAPPO code/pseudocode should also appear.	Add Appendix B: MAPPO Training Script, Appendix C: SUMO Route Files, Appendix D: YOLO Detection Script.
References	Some references are likely correct, but citation formatting must be checked.	Inconsistency in citation style reduces academic quality.	Use a consistent IEEE format. Ensure all figures adapted from papers or websites have proper citations.
Writing quality	The report is generally readable, but there are some grammar and spacing issues.	Technical writing should be polished before submission.	Correct phrases such as “a very strategically placed settlement”, “higher speeding” , and inconsistent capitalization of optimization/optimisation.

Major technical strength	The final MAPPO system is much stronger than a simple traffic-light simulation.	This should be highlighted properly.	Present the final contribution as: SUMO-based Palapye A1 corridor + YOLO sensing foundation + PPO baseline + MAPPO corridor controller.
Major technical weakness	The report still appears like two reports joined together: Semester 1 PPO and Semester 2 MAPPO.	The structure must be reorganized into one coherent final project.	Use this flow: Problem → Literature → Methodology → Single-agent baseline → MAPPO extension → Comparative results → Discussion → Conclusion.
Final submission readiness	Technically promising but not yet polished.	The project is strong, but the report needs cleanup before final assessment.	Priority fixes: restructure Chapters 4–7, remove duplicated tables, justify synthetic data assumptions, provide exact MAPPO numerical results, clarify YOLO non-integration, and correct chapter numbering.